

Program	Bachelor of Computer Applications (DS & AI)				
Year	II	Semester		III	
Course Name	Information & Data Security				
Code	BCADSN23211				
Course Type	GE	L	T	P	Credit
Pre-Requisite		2	1	0	3
Course Objectives	In this course, student will systematically study the fundamental principles of computer system security, including access control, security policies, software vulnerabilities, web security and various authentication mechanisms.				
Course Outcomes					
CO1	To understand the basics of information security.				
CO2	To learn about how to maintain the information and data security i.e., confidentiality, integrity and availability.				
CO3	Understanding the basic concept of security policies.				
CO4	The student will be able to understand the basics of security, policies, cryptographic algorithms, and its issues along with its countermeasures				
Module	Course Contents			Contact Hrs.	Mapped CO
1	Introduction to Information Security: Principles, CIA (Confidentiality, Integrity, Availability), Aspects of Information Security, Need for Security, Goals of Information Security, Features of a Good Security Policy, Security Attacks, Virus, DoS, Worms, Spyware, Ransomware, Security Services and Mechanisms, Security Standards.			12	CO1
2	Principles of Security: Steganography, Cryptographic Techniques: Plain Text and Cipher Text, Substitution Techniques, Types of Substitution Techniques, Transposition Techniques, Types of Transposition Techniques, Block Cipher Principles, Block Cipher Modes of Operation, Encryption and Decryption, Data Encryption Standard (DES) Algorithm, Strength of DES.			12	CO2
3	Introduction to Security Policies: Confidentiality, Integrity, Availability and Hybrid Policies, Academic Computer Security Policy: General University Policies, Information Risk Management, Risk Mitigation, Risk Handling Strategies and Risk Assessment, Information Classification – Guidelines, Types, Criteria for data Classification, Data Classification procedures, Classification Controls.			11	CO3
4	Authentication: Basics of Authentication, One Factor Authentication, Two Factor Authentication, Multi Factor Authentication, Passwords: Attacking a Password System, Countering Password Guessing, Biometrics: Fingerprints, Faces, Voices, Eyes and Combinations, Access Control, Types of Access Control.			10	CO4